EXACT DYADIC RECTANGLE DISCREPANCY AT A CRITICAL JORDAN RESONANCE

ANONYMOUS

Abstract. We give an explicit primitive, aperiodic 2×2 block substitution whose incidence matrix has a size-two Jordan block at the critical eigenvalue 2, equal to the linear expansion. A radius-one finite certificate recognizes the complete supertile grid. For the centered observable $f = \mathbf{1}_A - \mathbf{1}_B$, every translated dyadic rectangle of side lengths 2^m and 2^n satisfies a sharp discrepancy law:

$$\sup_{\substack{x \in X_\vartheta \\ t \in \mathbb{Z}^2}} \left| \sum_{q \in t + [0,2^m) \times [0,2^n)} f(x_q) \right| = 2^{\max(m,n)} \left(1 + \frac{\min(m,n)}{2} \right).$$

The proof combines an exact generalized-eigenvector identity with a pointwise periodic envelope. Thus the critical Jordan logarithm is not only visible on complete supertiles: its sharp worst-case coefficient is determined uniformly over translations and for every dyadic aspect ratio.

1. Introduction

For primitive substitution systems, incidence eigenvalues govern deviations of letter counts from their invariant frequencies. This principle is classical in one-dimensional symbolic discrepancy [1] and has far-reaching higher-dimensional tiling forms [2]. At a boundary eigenvalue—modulus equal to the linear boundary scaling—a nontrivial Jordan block can introduce logarithmic losses. General theory gives robust bounds, but it need not determine the sharp coefficient for a concrete symbolic window.

We exhibit a three-letter planar block substitution for which the critical mechanism can be computed exactly. Two finite structures cooperate. First, the incidence matrix has a length-two left Jordan chain at eigenvalue 2. Second, the three level-n observable arrays admit pointwise lower and upper envelopes that are 2^n-periodic. A translated 2^n square samples each residue once, turning those pointwise envelopes into sharp sum bounds.

The recognizability assertion is also explicit. The 20 legal 2×2 patches form a substitution-closed set, while the 63 legal 3×3 patches split into four disjoint phase classes. This is a complete radius-one certificate, rather than an appeal to recognizability in the abstract. General recognizability and its computability for multidimensional constant-shape substitutions have been developed much more broadly [3–5]; those general results are not claimed here.

2. The substitution and its subshift

Let ϑ be the 2×2 substitution on $\{A, B, C\}$ given by

$$\vartheta(A) = \begin{pmatrix} A & A \\ B & C \end{pmatrix}, \qquad \vartheta(B) = \begin{pmatrix} B & B \\ B & C \end{pmatrix}, \qquad \vartheta(C) = \begin{pmatrix} A & C \\ C & C \end{pmatrix}. \tag{1}$$

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Let $X_\vartheta \subset \{A,B,C\}^{\mathbb{Z}^2}$ be the subshift whose finite patches occur in iterates $\vartheta^n(a)$. Rows index output letters and columns index parent letters in the incidence matrix

$$M = \begin{pmatrix} 2 & 0 & 1 \\ 1 & 3 & 0 \\ 1 & 1 & 3 \end{pmatrix}. \tag{2}$$

Since

$$M^2 = \begin{pmatrix} 5 & 1 & 5 \\ 5 & 9 & 1 \\ 6 & 6 & 10 \end{pmatrix} > 0,$$

the substitution is primitive. Its Perron eigenvalue is 4, and its normalized right Perron vector is

$$\nu = (1/4, 1/4, 1/2)^\mathsf{T}. \tag{3}$$

2.1. **A finite recognizability certificate.** Write $\mathcal{L}_2$ for the set of legal 2×2 patches. Starting with the three images in (1), repeatedly adjoin every 2×2 subpatch of $\vartheta(P)$ for every current patch P. The cardinalities are

$$3 \longrightarrow 13 \longrightarrow 20 \longrightarrow 20.$$

Thus the resulting 20-patch set is substitution-closed. Every fine 2×2 patch crosses at most a 2×2 parent patch, so induction shows that this fixed set is exactly $\mathcal{L}_2$.

Every legal 3×3 patch is contained in $\vartheta(P)$ for some $P \in \mathcal{L}_2$. Classify it by the location of its upper-left corner modulo $2\mathbb{Z}^2$. With one choice of origin, the class sizes in phase order $(0,0), (0,1), (1,0), (1,1)$ are

$$18, \quad 20, \quad 15, \quad 10, \tag{4}$$

and the four classes are pairwise disjoint. The accompanying script performs this exhaustive set calculation with exact strings.

Proposition 2.1 (radius-one recognizability)**.** *Every 3×3 patch in X_ϑ determines its phase modulo $2\mathbb{Z}^2$. Consequently every $x \in X_\vartheta$ has a unique level-one phase–parent pair: a unique offset $\tau \in \{0,1\}^2$ for the supertile grid and a unique parent configuration. Iterating gives a unique level-n cutting for every n.*

Proof. Take substituted approximants containing larger and larger centered patches of x. Pass to a subsequence with a fixed grid offset and then take a compact limit of their parent configurations; this gives a level-one cutting. If its offset is τ, a 3×3 window with upper-left corner z belongs to the finite phase class indexed by $(z-\tau) \bmod 2\mathbb{Z}^2$. For a second cutting with offset τ', the same window would belong to the class indexed by $(z-\tau') \bmod 2\mathbb{Z}^2$. Pairwise disjointness of the four classes forces $\tau = \tau'$. At each common anchor, the corresponding 2×2 block is one of the three distinct images in (1), so it uniquely determines the parent letter. This proves uniqueness at level one, and iteration proves the general case. □

Corollary 2.2. *The subshift X_ϑ is aperiodic, minimal, and uniquely ergodic, with letter frequencies* (3)*.*

Proof. If $p \in \mathbb{Z}^2$ is a period, the intrinsic phase detector forces $p \in 2\mathbb{Z}^2$. Desubstitution then makes $p/2$ a period. Iteration forces p to be divisible by 2^n for every n, hence $p = 0$. Primitivity gives minimality and uniform patch frequencies, and therefore unique ergodicity; the letter frequencies are the normalized Perron vector. □

3. The critical Jordan chain

Let

$$f(A)=1, \qquad f(B)=-1, \qquad f(C)=0.$$

By (3), f has invariant mean zero. Define the left row vectors

$$w_1=(1,-1,0), \qquad w_0=(-1,-1,1).$$

Direct multiplication gives

$$w_0M=2w_0, \qquad w_1M=2w_1+w_0, \tag{5}$$

while

$$\det(\lambda I-M)=(\lambda-4)(\lambda-2)^2.$$

Thus eigenvalue 2 has a size-two Jordan block. From (5),

$$w_1M^n=2^nw_1+n2^{n-1}w_0. \tag{6}$$

For $a\in\{A,B,C\}$ let $S_n(a)$ be the sum of f over the complete n-supertile $\vartheta^n(a)$. Evaluating (6) on the three coordinate vectors gives the exact identities

$$\begin{aligned} S_n(A)&=2^n\left(1-\frac{n}{2}\right),\\ S_n(B)&=-2^n\left(1+\frac{n}{2}\right),\\ S_n(C)&=n2^{n-1}. \end{aligned} \tag{7}$$

The eigenvalue modulus 2 equals the linear expansion, hence the designation *critical.* The Jordan factor n becomes $\log_2 R$ at side length $R=2^n$.

4. A periodic-envelope lemma

Let F_n^a be the $2^n\times 2^n$ array obtained by applying f entrywise to $\vartheta^n(a)$. The block recursions from (1) are

$$F_{n+1}^A=\begin{pmatrix}F_n^A & F_n^A\\ F_n^B & F_n^C\end{pmatrix}, \quad F_{n+1}^B=\begin{pmatrix}F_n^B & F_n^B\\ F_n^B & F_n^C\end{pmatrix}, \quad F_{n+1}^C=\begin{pmatrix}F_n^A & F_n^C\\ F_n^C & F_n^C\end{pmatrix}.$$

Lemma 4.1 (pointwise envelope)**.** *For every $n\ge 0$,*

$$F_n^B\le F_n^A, \qquad F_n^B\le F_n^C$$

entrywise. Moreover $F_n^A\le F_n^C$ except at the single position $p_n=(0,2^n-1)$, where $F_n^A(p_n)-F_n^C(p_n)=1$.

Proof. The assertion is immediate at $n=0$. Substitute the three displayed block recursions. Each required comparison reduces to the induction hypothesis in one of the four blocks. The unique exceptional position moves into the upper-right block, from p_n to $p_{n+1}=(0,2^{n+1}-1)$. ∎

Extend F_n^B periodically to $\mathbb{Z}^2$, and likewise extend

$$U_n=F_n^C+\mathbf{1}_{\{p_n\}}.$$

By unique level-n cutting and lemma 4.1, after shifting their common phase these arrays bound the entire observable field of every $x\in X_\vartheta$ pointwise. Every square $Q=t+[0,2^n)^2$, $t\in\mathbb{Z}^2$, samples each residue class modulo $2^n\mathbb{Z}^2$ exactly once. Therefore

$$-2^n\left(1+\frac{n}{2}\right)\le\sum_{q\in Q}f(x_q)\le n2^{n-1}+1. \tag{8}$$

The left endpoint is attained by a complete B-supertile.

5. Exact discrepancy on every dyadic rectangle

For $m, n \geq 0$, define

$$D(m,n) = \sup \left\{ \left| \sum_{q \in t+[0,2^m)\times[0,2^n)} f(x_q) \right| : x \in X_\vartheta,\ t \in \mathbb{Z}^2 \right\}.$$

Theorem 5.1 (sharp dyadic rectangle law)**.** *For all $m, n \geq 0$,*

$$D(m,n) = 2^{\max(m,n)} \left(1 + \frac{\min(m,n)}{2}\right). \tag{9}$$

Proof. First take $m = n = k$. The absolute value of the lower endpoint in (8) dominates the upper endpoint, and a complete B-supertile attains it. Hence $D(k,k) = 2^k(1 + k/2)$.

Suppose $m = k \leq n = \ell$. Partition a $2^k \times 2^\ell$ rectangle into $2^{\ell-k}$ dyadic squares of side 2^k. The triangle inequality and the square case give

$$D(k,\ell) \leq 2^{\ell-k} D(k,k) = 2^\ell(1 + k/2).$$

For equality, take a complete ℓ-supertile of type B and its top strip of height 2^k. The top row of the parent array $\vartheta^{\ell-k}(B)$ consists entirely of B's, so the strip is a row of $2^{\ell-k}$ complete k-supertiles of type B and attains the negative bound. If $n = k \leq m = \ell$, use the left strip instead; the left column of $\vartheta^{\ell-k}(B)$ also consists entirely of B's. □

In particular, square windows of side $R = 2^n$ have exact worst-case discrepancy

$$D(n,n) = R\left(1 + \frac{1}{2}\log_2 R\right).$$

The logarithm is therefore a global translated-window phenomenon, not merely an ancestral-supertile calculation.

6. Scope and reproducibility

The finite control checks primitivity, the Jordan identities, the supertile formulas through level 32, and the pointwise envelope through level 8. Those level-bounded tests are regression checks for the displayed algebra and induction. Separately, starting from the three substitution images, the script constructs the complete 2×2 patch closure $3 \to 13 \to 20 \to 20$ and then all 63 legal 3×3 patches, verifying that their four phase classes are pairwise disjoint. This patch-language calculation is the exhaustive finite base of the recognizability proof.

The result should not be read as a general discrepancy theorem. General spectral control of substitution discrepancy belongs to the existing theory [1, 2]. In the planar critical size-two-Jordan regime, the general Lipschitz-domain estimate of Bufetov and Solomyak [2, Corollary 4.5] allows $O(R(\log R)^2)$; the special pointwise order in (1) is what yields the exact $R \log R$ law here. We claim only this explicit system, its radius-one certificate, its exact Jordan chain, and formula (9). External priority and release questions remain outside this internal draft.

References

[1] Boris Adamczewski. Symbolic discrepancy and self-similar dynamics. *Annales de l'Institut Fourier*, 54(7):2201–2234, 2004. doi: 10.5802/aif.2079. URL https://www.numdam.org/articles/10.5802/aif.2079/.

[2] Alexander I. Bufetov and Boris Solomyak. Limit theorems for self-similar tilings. *Communications in Mathematical Physics*, 319(3):761–789, 2013. doi: 10.1007/s00220-012-1624-7. URL https://doi.org/10.1007/s00220-012-1624-7.

[3] Christopher Cabezas. Homomorphisms between multidimensional constant-shape substitutions. *Groups, Geometry, and Dynamics*, 17(4):1259–1323, 2023. doi: 10.4171/GGD/726. URL https://doi.org/10.4171/GGD/726.

[4] Christopher Cabezas and Julien Leroy. Decidability of the isomorphism problem between multidimensional substitutive subshifts. *Ergodic Theory and Dynamical Systems*, 45(7):2054–2094, 2025. doi: 10.1017/etds.2024.87. URL https://doi.org/10.1017/etds.2024.87.

[5] Boris Solomyak. Nonperiodicity implies unique composition for self-similar translationally finite tilings. *Discrete and Computational Geometry*, 20(2):265–279, 1998. doi: 10.1007/PL00009386. URL https://doi.org/10.1007/PL00009386.